

Mycelium Extraction Framework: Development & Testing Report

Author: McLean Taggart **Date:** March 24, 2026 **Status:** Framework validated, ready for pilot extraction

1. Executive Summary

This session formalized the Mycelium claim extraction architecture, built a strict Pydantic schema and extraction prompt, and tested extraction on a real protein-interaction paper using three Claude models (Opus, Sonnet, Haiku). Key outcomes:

- **Schema created:** 880-line Pydantic schema defining `ExtractionResult` with `AssertionDraft`, `EvidenceUnit`, and `CitationContext` models, 12 `StrEnum` types
 - **Architectural decision:** Background claims replaced with `CitationContext` signals — edges emerge from cross-paper semantic comparison and explicit citation signals, not from duplicating prior claims as assertions
 - **Model recommendation:** Sonnet as primary extraction model (100% scope completeness, rich citation contexts, best cost-quality balance)
 - **Cost estimate:** ~\$10K (Haiku Batch) to ~\$19K (two-tier) for full bioRxiv corpus (~320K papers)
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2. Architecture: Three-Layer Pipeline

The extraction framework sits in Layer 1 of a three-layer pipeline:

- **Layer 1 — Extraction** (this session): LLM processes a single paper → outputs `ExtractionResult`
- **Layer 2 — Projection** (future): Resolves claim identity across papers, detects edges, produces canonical `Assertion` nodes
- **Layer 3 — Graph** (future): PostgreSQL (metadata, embeddings) + Neo4j (graph traversal, confidence propagation)

What the LLM Extracts Per Paper

Component	Purpose	Count (typical)
AssertionDraft	Novel scientific claims at the mechanism level	11–22 per paper
EvidenceUnit	Distinct experimental results with full statistical detail	16–23 per paper
CitationContext	How this paper references specific prior findings (edge signals)	4–12 per paper
PaperProvenance	Paper-level metadata for source independence computation	1 per paper

The CitationContext Decision

In v1, background claims were extracted as full assertion drafts (`is_primary=false`). This inflated the graph with second-hand interpretations of claims that already exist as primary findings from their source papers.

v2 approach: Background claims are replaced with lightweight CitationContext signals that capture the exact citing sentence, what the author says the cited paper found, the relationship type (supports, contradicts, extends, qualifies, contextualizes, replicates, mentions), and links to this paper's novel assertions. This gives Layer 2 high-confidence edge candidates without node pollution.

Edges emerge two ways:

1. **Explicit:** Citation context signals (high precision, author-stated)
2. **Implicit:** Semantic comparison across papers' novel assertions (high recall, computationally discovered)

3. Schema Design

The Pydantic schema (`src/mycelium/extraction_schema.py`) defines 12 enumerations and 16 models.

Enumerations

Enum	Values
AssertionTypeEnum	mechanistic_causal, correlational, comparative, existence, absence, conditional, methodological
CausalTypeEnum	necessary, sufficient, necessary_and_sufficient, contributory, modulatory
HedgeLevelEnum	high, medium, low
SectionEnum	abstract, introduction, methods, results, discussion, conclusion
FunctionEnum	novel_finding, background (QA only), replication, hypothesis, interpretation, limitation, methodological_note
EntityTypeEnum	gene, protein, chemical, disease, pathway, cell_type, organism, method, other
EvidenceStrengthEnum	systematic_review_meta_analysis → expert_opinion (8 levels)
ResultDirectionEnum	positive, negative, null_powered, null_underpowered, not_reported
CitationRelationshipEnum	supports, contradicts, extends, qualifies, contextualizes, replicates, mentions

Key Design Decisions

1. **Scope is first-class** — species, tissue, cell_type, disease, condition, developmental_stage, in_vitro are mandatory fields
 2. **Hedging preserved verbatim** — the author's exact language is captured alongside structured classification
 3. **Section context determines epistemic function** — same sentence means different things in Introduction vs Results
 4. **Lazy entity normalization** — surface forms at extraction time, ontology IDs populated during enrichment
 5. **strict=False** — Pydantic v2 strict mode breaks JSON validation for StrEnum; LLM output produces strings, not enum instances
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4. Test Paper

Title: Rai14 is a novel interactor of Invariant chain that regulates macropinocytosis **Authors:** Lobos Patorniti, Zulkefli, McAdam, Vargas, Bakke, Progida **Journal:** Frontiers in Immunology (2023) **DOI:** 10.3389/fimmu.2023.1182180

Selected because it is a primary research paper with clear protein-protein interaction claims (Rai14–Ii, Rai14–myosin II), multiple experimental methods (Y2H, co-IP, live imaging, knockdown, migration assays), quantitative results, and is open access.

5. Extraction Results: v1 (Opus vs Sonnet)

The first test compared Opus and Sonnet on the v1 schema, which included background assertions as full assertion drafts.

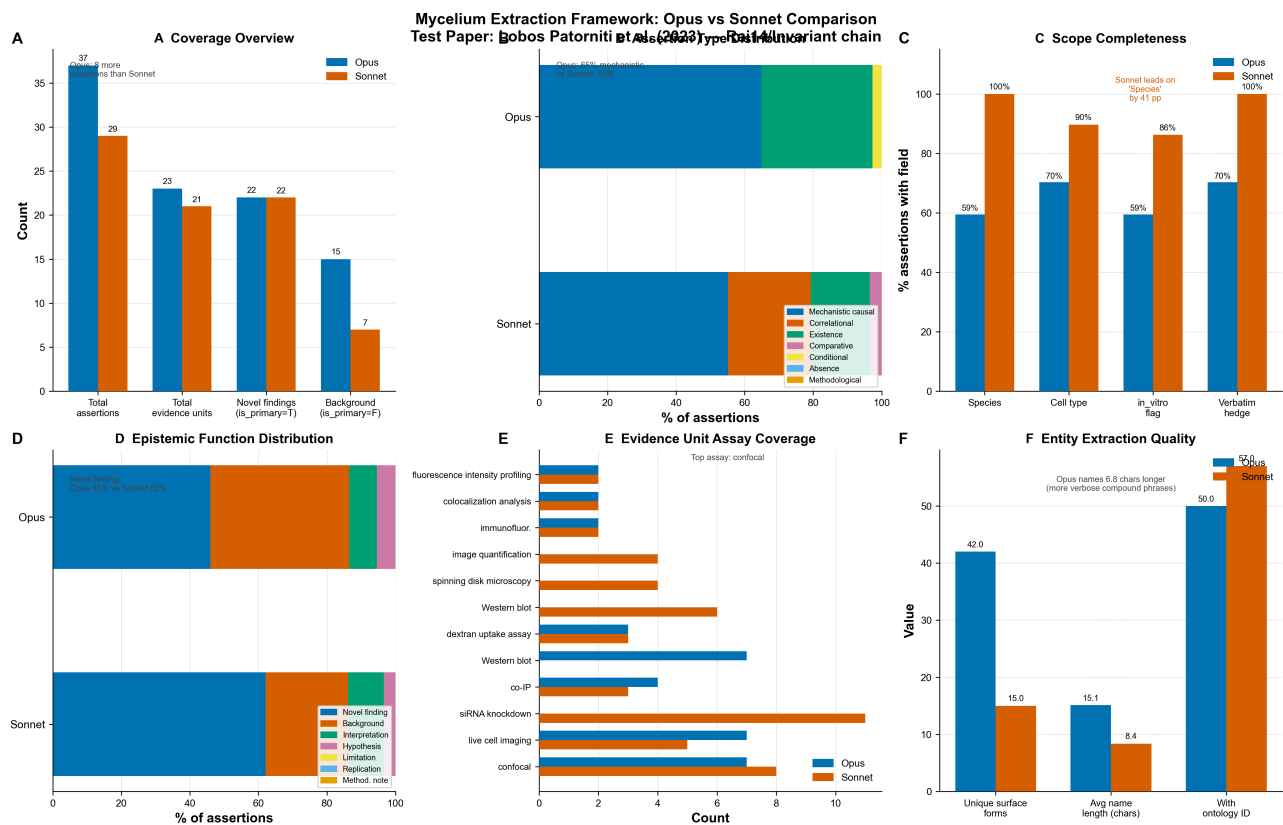


Figure 1: v1 Opus vs Sonnet comparison. Panel C (Scope Completeness) shows the most dramatic difference — Sonnet fills scope fields at ~100% vs Opus at ~60%.

Key Findings (v1)

Metric	Opus	Sonnet
Assertion drafts	37	29
Evidence units	23	21
Novel findings	22	22
Background claims	15	7
Scope completeness (species)	59%	100%
Hedging captured	70%	100%
Entity surface forms	42 (verbose)	15 (clean)
Cross-reference integrity	Perfect	Perfect

Opus extracted more assertions overall, but the advantage was entirely in background claims. Both models found the same 22 novel findings. Opus's entity extraction produced verbose compound phrases that would complicate entity resolution.

Sonnet had significantly better field discipline — 100% scope fill, 100% hedging capture. Every assertion knew what species, cell type, and experimental context it applied to.

6. Extraction Results: v2 (Three-Model Comparison)

The v2 schema replaced background assertions with CitationContext. We tested Sonnet and Haiku on the updated framework.

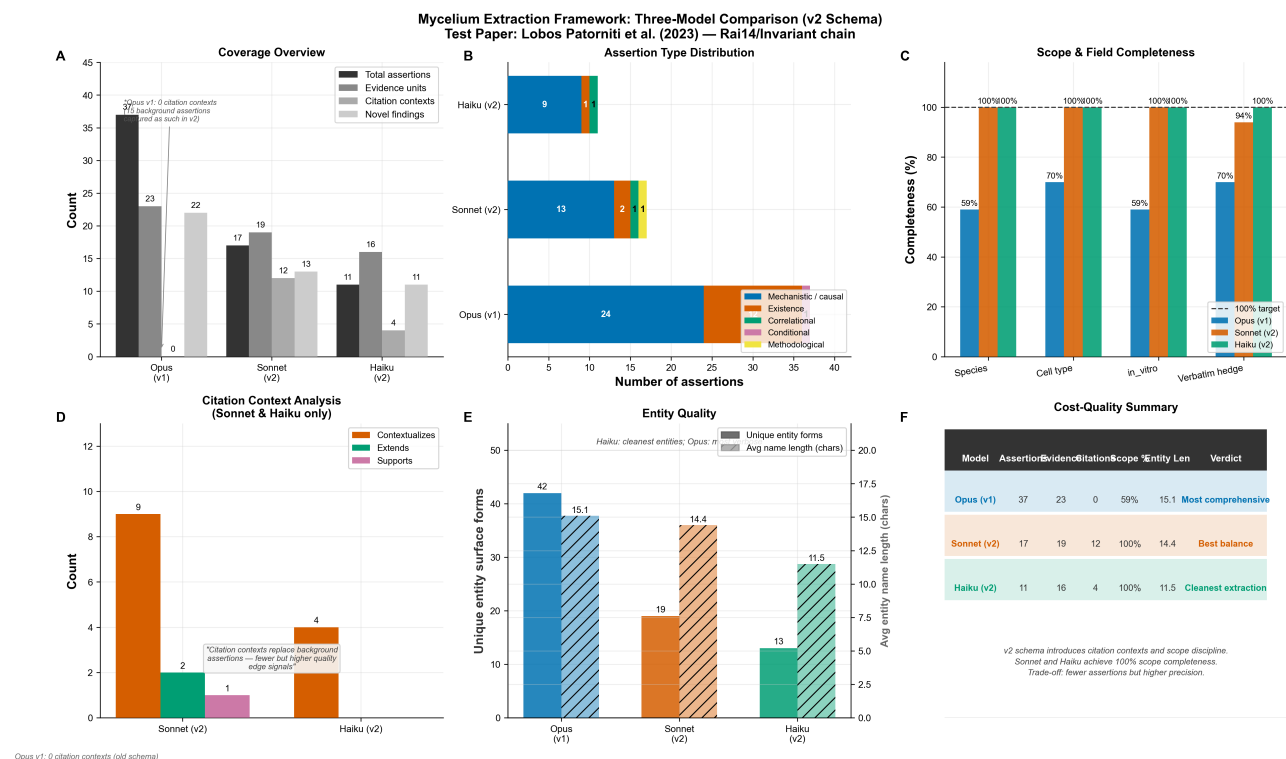


Figure 2: v2 three-model comparison. Panel C shows both Sonnet and Haiku achieve ~100% scope completeness. Panel D shows Sonnet extracts 3x more citation contexts with richer relationship diversity.

Three-Model Comparison

Metric	Opus (v1)	Sonnet (v2)	Haiku (v2)
Assertion drafts	37	17	11
Evidence units	23	19	16
Citation contexts	0 (old schema)	12	4
Novel findings	22	17	11
Background assertions	15	0	0
Scope completeness	59%	100%	100%
Avg entity name length	15.1 chars	14.4 chars	11.5 chars

Citation Context Quality

Sonnet extracted 3x more citation contexts than Haiku (12 vs 4), with richer relationship diversity:

- **Sonnet**: contextualizes (9), extends (2), supports (1)
- **Haiku**: contextualizes (4) only

Haiku captures basic contextual references but misses when a prior finding is being extended or supported.

7. Cost Analysis

Per-Paper Extraction Cost

Paper Length	Sonnet	Sonnet Batch	Haiku	Haiku Batch
Short (5 pg)	\$0.22	\$0.11	\$0.04	\$0.02
Medium (10 pg)	\$0.33	\$0.16	\$0.06	\$0.03
Typical (15 pg)	\$0.37	\$0.18	\$0.07	\$0.03
Long (25 pg)	\$0.56	\$0.28	\$0.10	\$0.05

Full bioRxiv Corpus (~320K papers)

Strategy	Cost	Notes
Sonnet Batch	\$53K–\$60K	Best quality
Haiku Batch	\$10K–\$11K	Cheapest viable
Two-tier (Haiku all + Sonnet top 20%)	~\$19K	Recommended
Fine-tuned 8B (future)	\$2K–\$5K inference	Requires \$20K+ training investment first

Fine-Tuning vs API

At bioRxiv scale (320K papers), the API is cheaper (\$10–19K) than fine-tuning (\$21–44K total). Fine-tuning only pays off at PubMed scale (36M papers), where API costs reach \$1M+ but a fine-tuned model runs for ~\$3K.

Input Format

bioRxiv provides full text through its API in structured format — no PDF parsing needed for the target corpus. The extraction results from our test (which used parsed text) are representative.

8. Framework Improvements Identified

1. **Add `biological_process` to `EntityTypeEnum`** — both models mapped macropinocytosis/cell migration to `pathway` as a workaround
 2. **Clarify `assertion_type` for binding evidence** — co-IP/pulldown = existence of interaction; knockdown + phenotype = mechanistic_causal
 3. **Enforce single-entity rule** — `subject_entity` and `object_entity` must be individual biological entities
 4. **Standardize figure references** — enforce consistent format (e.g., "Figure 1A")
 5. **Add prompt caching** — system prompt is ~7,650 tokens, identical across papers; caching saves ~90% on input cost
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9. Recommendations

For the Pilot (Next Steps)

1. Apply the 5 framework improvements to the schema and prompt
2. Run Sonnet v2 extraction on 50–100 papers from the PPI pilot domain to validate at scale
3. Begin gold standard annotation on the first 10 papers (McLean + 2 domain experts)
4. Build Layer 2 projection pipeline (claim identity resolution, edge detection)

For Production Extraction

- **Primary model:** Sonnet (best balance of coverage, scope discipline, citation context quality)
- **Corpus strategy:** Haiku Batch on full bioRxiv (\$10K), Sonnet re-extract on top 50K papers (\$9K)
- **Input format:** bioRxiv API provides structured text — no PDF parsing needed

For the Paper

- The extraction comparison (v1 vs v2, three models) is publishable as a methods evaluation
- The CitationContext decision is a novel contribution — most knowledge graphs extract all claims indiscriminately
- Calibration data: extraction quality by paper section, claim type, and method type

10. Files Produced

File	Description
src/mycelium/extraction_schema.py	Pydantic schema (880 lines, 12 enums, 16 models)
prompts/mycelium_extraction_prompt.md	Three-stage extraction prompt
outputs/extraction_test_opus.json	Opus v1 extraction (37 assertions, 23 EU)
outputs/extraction_test_sonnet_v2.json	Sonnet v2 extraction (17 assertions, 19 EU, 12 citations)
outputs/extraction_test_haiku.json	Haiku v2 extraction (11 assertions, 16 EU, 4 citations)
outputs/extraction_comparison.png	v1 comparison figure (6 panels)
outputs/extraction_comparison_v2.png	v2 three-model comparison figure (6 panels)
outputs/extraction_report_opus_v1.pdf	Full Opus extraction report
outputs/ extraction_report_sonnet_v2.pdf	Full Sonnet extraction report

File	Description
outputs/extraction_report_haiku_v2.pdf	Full Haiku extraction report